FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2019

TEST DATE: 16-12-2018

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – I

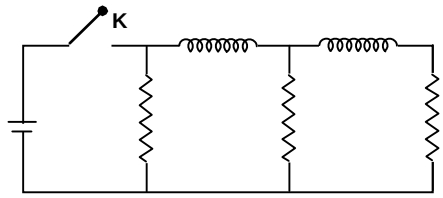
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two identical blocks of metal are at 20°C and 80°C respectively. The specific heat of the material of the two blocks increases with temperature. Which of the following is true about the final temperature T_f when the two blocks are brought into contact? (assuming that no heat is lost to the surroundings)
   (A) T_f will be 50°C
   (B) T_f will be more than 50°C
   (C) T_f will be less than 50°C
   (D) T_f can be either more than or less than 50°C depending on the precise variation of the specific heat with temperature.

2. An electron in an electron microscope with initial velocity $v_0\hat{i}$ enters a region of a transverse electric field $E_0\hat{j}$. The time taken for the change in its de-Broglie wavelength from the initial value of λ to $\lambda/3$ is proportional to
   (A) E_0
   (B) $1/E_0$
   (C) $\frac{1}{\sqrt{E_0}}$
   (D) $\sqrt{E_0}$

3. In the circuit shown below, all the inductors (assumed ideal) and resistors are identical. The current through the resistance on the right is I after the key K has been switched on for along time. The currents through the three resistors (in order, from left to right) immediately after the key is switched off are



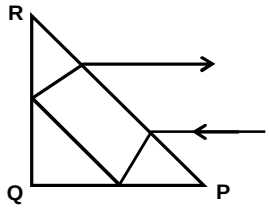
   (A) 2I upwards, I downwards and I downwards
   (B) 2I downwards, I downwards and I downwards
   (C) I upwards, I downwards and I downwards
   (D) 0, I downwards and I downwards

4. The moments of inertia (in kg-m^2) of a non-uniform circular disc of mass 1 kg and radius 25 cm about four mutually perpendicular tangents AB, BC, CD and DA are 2, 4, 6, 7 respectively (the square ABCD circumscribes the disc). The distance of the centre of mass (in m) of the disc from its geometrical centre is
   (A) 5
   (B) 4
   (C) 3
   (D) 2

5. A small box resting on one edge of the table is struck in such a way that it slides off the other edge, 1 m away, after 2 seconds. The coefficient of kinetic friction between the box and the table-
   (A) must be less than 0.05
   (B) must be exactly zero
   (C) must be more than 0.05

(D) must be exactly 0.05

6. A ray of light incident parallel to the base PQ of an isosceles right angled triangular prism PQR suffers two successive total internal reflections at the face PQ and QR before emerging reversed in direction as shown. If the refractive index of the material of the prism is μ, then



(A) $\mu > \sqrt{5}$

(B) $\sqrt{3} < \mu < \sqrt{5}$

(C) $\sqrt{2} < \mu < \sqrt{3}$

(D) $\mu > \sqrt{2}$

7. A rigid body in the shape of a "V" has two equal rams made of uniform rods with an angle θ between them. When the body is suspended from one end, the other arm is horizontal. The value of θ is

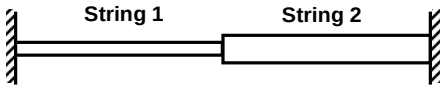
(A) $\cos^{-1}\left(\frac{1}{3}\right)$

(B) $\cos^{-1}\left(\frac{1}{2}\right)$

(C) $\cos^{-1}\left(\frac{1}{4}\right)$

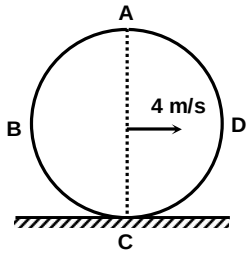
(D) $\cos^{-1}\left(\frac{1}{6}\right)$

8. A composite string is formed by joining two string of equal length but different mass per unit length and stretched between two rigid supports. The string 2 has mass per unit length 4 times the mass per unit length of string 1. A progressive wave, travelling in string 1 to the right, is incident on the boundary what fraction of incident energy goes in string 2?



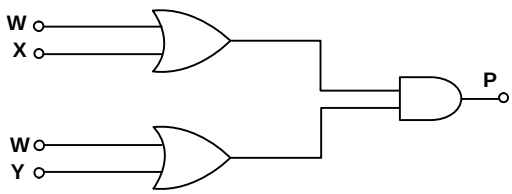
(A) 1/8
(B) 4/9
(C) 2/3
(D) 8/9

9. A sphere of mass 1 kg and radius 1 m rolls on a horizontal surface with a speed 4 m/s. The force by one half of the sphere ABC (line AC is vertical) on the other ADC is (in N)



(A) 1
(B) 2
(C) 3
(D) 4

10. The diagram of a logic circuit is given. The output of the circuit is represented by



(A) W.(X+Y)
(B) W.(X.Y)
(C) W+(X.Y)
(D) W+(X+Y)

11. The minimum speed of flow of water through a long cylindrical pipe of diameter 2 cm so that the flow becomes turbulent is (density of water = 1×10^3 kg/m^3, viscosity of water = 1×10^{-3} N-s/m^2, Reynold's number for critical velocity = 3000)
(A) 1.5 m/s
(B) 0.15 m/s
(C) 2 m/s
(D) 1 m/s

12. Two identical metallic sheets of area $\frac{10^3}{17 \times 27}$ m^2 are arranged parallel with some separation between them in vacuum. Thermal energy at a constant rate 'P' is generated in one of the sheets by passing current through it. In steady state, the temperature of the other sheet is found to be 300 K. The value of P (in kW) is $\left(\sigma = \frac{17}{3} \times 10^{-8} \frac{W}{m^2 K^4} \right)$
(A) 1
(B) 2
(C) 3
(D) 4

13. The lower end of uniform string whose upper end is attached to ceiling, is oscillated simple harmonically with a frequency f and amplitude A. The relation between A and f, if the maximum acceleration of the particle is equal to the acceleration of wave is (g = acceleration due to gravity)
(A) $\pi^2 f^2 A = g$
(B) $2\pi^2 f^2 A = g$
(C) $4\pi^2 f^2 A = g$
(D) $8\pi^2 f^2 A = g$

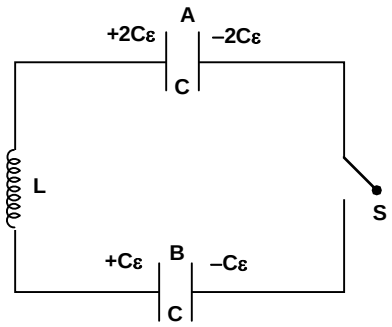
14. An eye can distinguish between two points on an object placed 25 cm from the eye if they are separated by more than 0.22 mm. A compound microscope of 20 D objective and 10 D eyepiece separated by 20 cm is used to view this object. The final image is formed at a distance of 25 cm from the eye. The minimum separation between two points which can now be distinguished (in mm) is nearly.
(A) 0.01
(B) 0.02
(C) 0.04
(D) 0.05

15. A sonometer wire of length 1.5 m is made of steel. The tension in it produces an elastic strain of 1%. If density and Young's modulus of steel are 7.7×10^3 kg/m^3 and 2.2×10^{11} N/m^2, the fundamental frequency of the wire (in Hz) is nearly
(A) 178
(B) 200
(C) 700
(D) 188

16. It is observed that output power from a (non-ideal) cell is same when the current through it is either 2A or 8A. The output power will be maximum for a current (in A) of
(A) 3
(B) 4
(C) 5
(D) 6

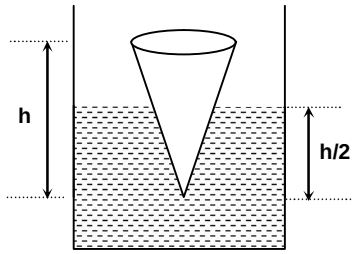
17. A gas enclosed in a vessel at an initial pressure of 144 cm of Hg, is being pumped out of the vessel by means of a piston-pump. The pressure reduces to 36 cm of Hg after two strokes. After another two strokes, the pressure (in cm of Hg) is
(A) 8
(B) 16
(C) 12
(D) 9

18. In the adjacent circuit, initially capacitors A and B are charged by voltage 2ε and ε respectively. Initial current through the inductor is zero and switch 'S' is closed at t = 0. The maximum current through the inductor is



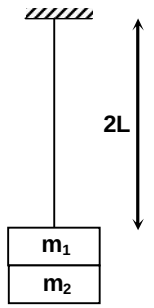
(A) $\varepsilon\sqrt{\frac{C}{L}}$
(B) $\varepsilon\sqrt{\frac{C}{2L}}$
(C) $\varepsilon\sqrt{\frac{2C}{L}}$
(D) $\varepsilon\sqrt{\frac{C}{3L}}$

19. A solid cone of height h with heavy mass attached to it floats upside down in water, as shown in the figure. In equilibrium, water reaches a height h/2. When displaced downward slightly and released, it performs simple harmonic motion. The frequency of these oscillations is

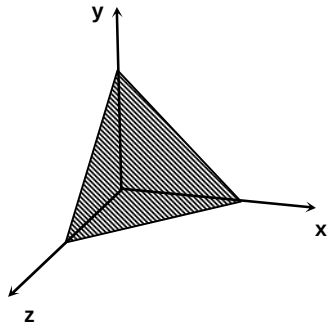


(A) $\frac{1}{2\pi}\sqrt{\frac{6g}{h}}$
(B) $\frac{1}{2\pi}\sqrt{\frac{g}{h}}$
(C) $\frac{1}{2\pi}\sqrt{\frac{3g}{h}}$
(D) $\frac{1}{2\pi}\sqrt{\frac{8g}{h}}$

20. From a fixed support O, hangs an elastic string of negligible mass and natural length L. Masses m_1 and m_2 attached at the lower end, elongate the string to 2L. When m_2 is removed gently, m_1 is able to bounce back to O. The ratio m_2/m_1 is (assume that the string does not obstruct the motion of m_1)
(A) 1
(B) 2
(C) 3
(D) 4



21. A point charge Q is placed at $\left(0,\frac{a}{\sqrt{2}},0\right)$. Consider a triangular surface having vertices at (a, 0, 0), (0, a, 0) and (0, 0, a). The flux of the electric field through this surface is



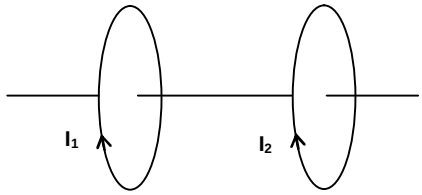
(A) $\frac{Q}{6\varepsilon_0}$

(B) $\frac{7}{21}\frac{Q}{\varepsilon_0}$

(C) $\frac{3}{25}\frac{Q}{\varepsilon_0}$

(D) $\frac{5}{24}\frac{Q}{\varepsilon_0}$

22. A system consists of two coaxial circular current carrying rings as shown in the figure. Their self inductances are L_1 and L_2 respectively and M is the coefficient of mutual induction. They carry current in the indicated directions. The magnetic energy of the system is



(A) $\frac{1}{2}L_1i_1^2+\frac{1}{2}L_2i_2^2-Mi_1i_2$

(B) $\frac{1}{2}L_1i_1^2+\frac{1}{2}L_2i_2^2+Mi_1i_2$

(C) $\frac{1}{2}L_1i_1^2+\frac{1}{2}L_2i_2^2-\frac{1}{2}Mi_1i_2$

(D) $\frac{1}{2}L_1i_1^2+\frac{1}{2}L_2i_2^2+\frac{1}{2}Mi_1i_2$

23. The main scale of vernier callipers reads in mm and its vernier scale is divided into 8 divisions which coincide with 5 divisions of the main scale. When the two jaws touch each other, the zero of the vernier coincides with the zero of main scale. When a rod is tightly placed along its length between the jaws, it is observed that the zero of vernier scale lies just left to 36th division of main scale and fourth division of vernier scale coinides with one main scale division. The measured value is (in cm)

(A) 3.66
(B) 3.55
(C) 3.65
(D) 3.56

24. The number n of β particles emitted with energy E is given by $n \propto (E_{max}-E)^2 \times \sqrt{E}$, where E_{max} is the maximum energy released in β decay. The maximum number of antineutrinos would come out with energy approximately equal to

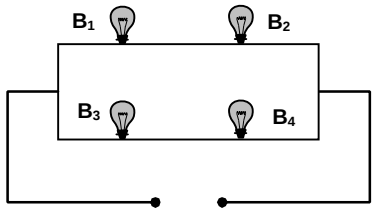
(A) $\frac{E_{max}}{5}$

(B) $\frac{4E_{max}}{5}$

(C) E_{max}

(D) zero

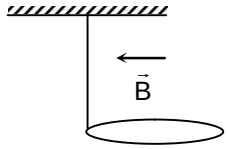
25. Four bulb are connected as shown in the figure. Their ratings are:



Bulb	Rating
1	24 W, 6 Ω
2	48 W, 3 Ω
3	72 W, 2 Ω
4	72 W, 2 Ω

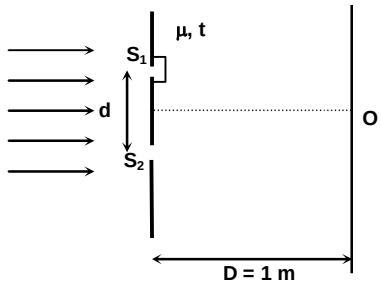
The maximum power produced by the combination is
(A) 216 W
(B) 208 W
(C) 307 W
(D) 117 W

26. A circular ring of radius 1 m and mass π kg suspended from the ceiling with the help of an ideal string, carries a current I. In the presence of a uniform horizontal magnetic field of 1 T, the ring remains in equilibrium with its plane horizontal. The current I is



(A) 10 A is ACW direction when viewed from above.
(B) 10 A is CW direction when viewed from above.
(C) 5 A is CW direction when viewed from above.
(D) 5 A is ACW direction when viewed from above.

27. A coherent parallel beam of light of wavelength λ is incident normally on the plane of slits. Slit S_1 is covered by a thin slab of refractive index $\mu = 1.5$ and thickness $t = 1 \times 10^{-5}$ m. The separation between the slits is d = 1 mm. The distance of the central maxima from point O is

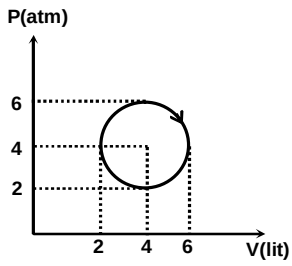


(A) 0.5 cm
(B) 1 cm
(C) 1.5 cm
(D) 2 cm

28. The maximum and minimum amplitudes of an AM wave are 10 V and 3V respectively. The modulation index is
(A) 0.3
(B) 10/3
(C) 13/30
(D) 7/13

29. The dominant mechanisms for motion of charge carriers in forward and reverse biased silicon p-n junctions are
(A) drift in forward biased, diffusion in reverse biased.
(B) diffusion in forward biased, drift in reverse biased.
(C) diffusion in both forward and reverse biased.
(D) drift in both forward and reverse biased.

30. One mole of an ideal gas undergoes a circular cycle centered at 4 atm, 4 lit as shown in the diagram. The maximum temperature attained in this process is close to



(A) 30/R
(B) 36/R
(C) 24/R
(D) 16/R

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. In ccp structure of mixed oxides, the lattice is made up of oxide ions. Tetrahedral voids are occupied by divalent X^{2+} ions and octahedral voids are occupied by trivalent Y^{3+} ions. If only 12.5% of tetrahedral voids are occupied, then the fraction of octahedral voids occupied is:
(A) 0.75
(B) 0.50
(C) 0.40
(D) 0.25

32. $\text{Cyclopentane} \xrightarrow[(1\,\text{eqv.})]{Cl_2, h\nu} [X] \xrightarrow[C_2H_5OH, \Delta]{KOH} [Y] \xrightarrow[CS_2]{O_3/Zn} [Z]$
Here [Z] is
(A) Pentanal
(B) Pentan-2-one
(C) Pentane-3-one
(D) Pentane 1, 5, dial

33. Which of the statement is correct?
(A) Li, when burnt in air, gives monoxide, peroxide and superoxide
(B) On heating Li_2CO_3, Li_2O and CO_2 are formed
(C) Among alkali metals, Li metal has high I.E. and impart no colour to flame
(D) Li_2SO_4 forms alum

34. In which of the following reaction XeF_4 is not reduced to Xe?
(A) $XeF_4 + O_2F_2$
(B) $XeF_4 + H_2$
(C) $XeF_4 + KI$
(D) $XeF_4 + SF_4$

35. Which of the following is incorrectly matched?
(A) $\left[Al(OH)_4\right]^-(aq) + NH_4^+(aq) \xrightarrow{\text{Slightly heat}} \text{white precipitate} + \text{liberation of } NH_3$
(B) $Pb^{2+}(aq) + 2Br^-(aq) \longrightarrow \text{Red precipitate}$
(C) $BiI_3(\text{black precipitate}) + H_2O(\ell) \xrightarrow{\Delta} \text{orange precipitate}$
(D) $S^{2-} + Na_2\left[Fe(CN)_5NO\right] \longrightarrow \text{Purple colouration}$

36. pH range of a basic indicator (InOH) is 3.3 – 4.7. What is the minimum value of $\left[In^+\right]/[InOH]$ above which the solution assumes only the colour of In^+ ion?
(A) 4
(B) 5
(C) 6
(D) 7

37. 200 cm^3 of 0.50 M $H_2C_2O_4$ solution was shaken with 9.0 g of activated charcoal for some time. 10 cm^3 of filtrate required 32 cm^3 of 0.25 M NaOH for just neutralization. The amount of adsorption of acid per gram of adsorbent is.
(A) 2 g
(B) 0.2 g
(C) 7.2 g
(D) 1.8 g

38. The reduction potential at 25°C for Fe^{3+}/Fe^{2+} electrode is 0.718 V. If $E^{o}_{Fe^{2+}/Fe} = -0.44$ V and $E^{o}_{Fe^{3+}/Fe} = -0.04$ V the ratio of molar concentration of Fe^{2+} ion to that of Fe^{3+} ion in solution is $\left[\frac{2.303RT}{F} = 0.06, \log 5 = 0.7\right]$
(A) 5
(B) 0.5
(C) 0.005
(D) 0.05

39. In Kjeldahl's method the gas evolved from 1.731 g sample of a fertilizer is passed into 50 ml of 0.20 N of H_2SO_4. 25 ml of 0.19 N NaOH are required for the titration of unused acid. The percentage of nitrogen by mass in fertilizer is:
(A) 4.2
(B) 5.25
(C) 4.75
(D) 6.8

40. What volume of NH_3 gas at NTP will be required to be passed into 30 ml of 1 N H_2SO_4 to bring down the acid normality to 0.2 N.
(A) 408 mL
(B) 53.76 mL
(C) 537.6 mL
(D) 40.8 mL

41. 24.0 g of diborane is treated with excess of ammonia at low temperature. The addition product so obtained is heated to 200°C and then cooled to room temperature. The colourless liquid so obtained weighs…..(Consider atomic mass of Boron = 11)
(A) 81 g
(B) 69.43 g
(C) 8. 5 g
(D) 46.29 g

42. Tetrahedral and colourless complexes among the following are:
(A) $K_2[Ni(CN)_4]$
(B) $Na_2[NiCl_4]$
(C) $K_2[Zn(CN)_4]$
(D) $Ni(CO)_4$

43. For the coagulation of a particular colloidal solution, the order of coagulation values of some electrolytes is in the order : $AlCl_3 > Na_2SO_4 > Na_3PO_4$
Which of the following conclusions is incorrect for such a colloidal solution?

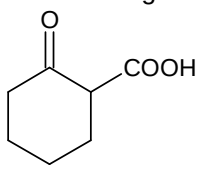
(A) Counter ions in the electrical double layer around the colloidal particles are negatively charged.
(B) It may be basic dye sol such as methylene blue.
(C) In electrophoresis, the sol particles move towards the anode.
(D) In electrosmosis, the dispersion medium moves towards the anode.

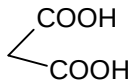
44. Consider the following reactions:
(i) $NH_4NO_3 \xrightarrow{\Delta}$ Products
(ii) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ Products
(iii) $Hg(NO_3)_2 \xrightarrow{\Delta}$ Products
(iv) $Na(NH_4)HPO_4,4H_2O \xrightarrow{\Delta}$ Products
What is the sum of the oxidation states of nitrogen 'N' in the compounds (product) formed in the above reactions?
(A) 2
(B) 6
(C) 5
(D) 3

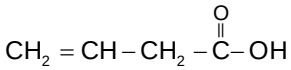
45. If 10^{-x} is the factor by which solubility of AgBr(s) [$K_{sp} = 10^{-10}$] changes with respect to initial solubility, when it is dissolved in 0.01 M NaBr (aq) solution. The value of 'X' is:
(A) 2
(B) 3
(C) 4
(D) 5

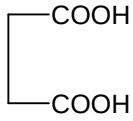
46. 'X' is one of substance formed when SO_2 and PCl_5 react together. It is a fuming liquid which dissolves in water to give a pungent smelling aqueous solution which turns blue litmus red. The "fuming liquid" is
(A) SO_2Cl_2
(B) HCl
(C) $SOCl_2$
(D) SO_2

47. Which of the following is not addition Homopolymer?
(A) Polystyrene
(B) Teflon
(C) Melamine
(D) Polyacrylonitrile

48. Which of the following compounds does not decarboxylate on heating?
(A) 

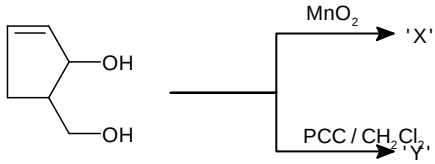
(B) 

(C) 

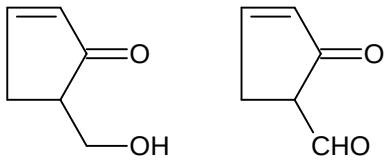
(D) 

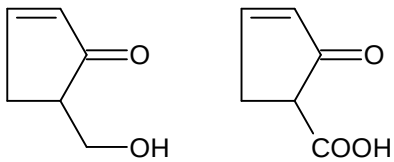
49. Which of the following statement is correct?
(A) Anglesite is a sulphate ore with +4 oxidation state of the metal
(B) Corundum is a sulphide ore with +3 oxidation state of the metal
(C) Argentite is an oxide ore with +1 oxidation state of the metal
(D) Siderite is a carbonate ore with +2 oxidation state of the metal

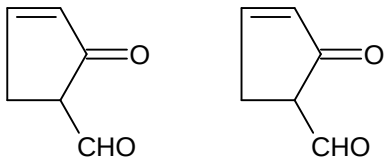
50. Which of the following statements is correct?
(A) Vitamin C and D are water soluble
(B) Vitamin B_6 is known as pyridoxine
(C) Morphine is used as antipyretic
(D) Vitamin E is also known as cyanocobalamin

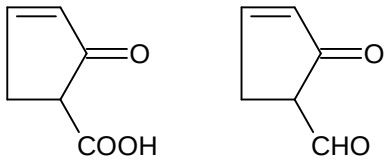
51. 

The compounds 'X' and 'Y' respectively are

(A) 

(B) 

(C) 

(D) 

52. For H-atom, as the orbit number increases, the distance between two successive orbits (r_1 = radius of first orbit)

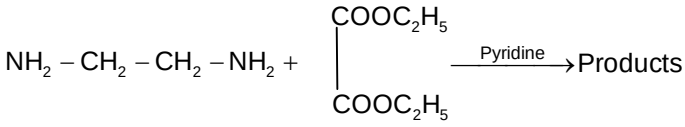
(A) Increases by $2r_1$

(B) increases by $(2n + 1)r_1$, where n is higher orbit

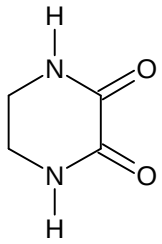
(C) increases by $(2n + 1)r_1$, where n is lower orbit

(D) remains constant

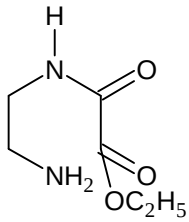
53. The major product of the following reaction is:



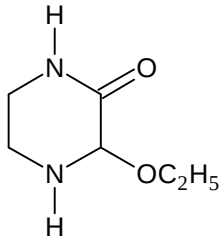
(A)



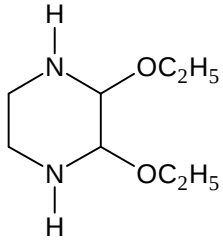
(B)



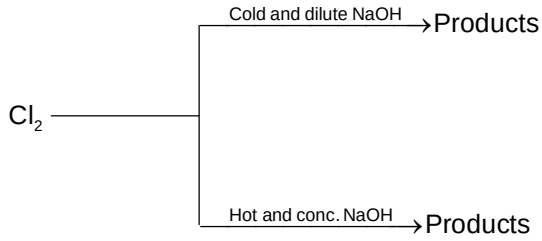
(C)



(D)



54.



The product which is not common in (i.e. present in one but not in the other) both the reactions is:

(A) NaOCl

(B) NaCl

(C) H_2O

(D) $NaClO_2$

55. For a first order reaction in gaseous phase, $A \longrightarrow B + C + D$, starting with initial pressure 400 mm. If the total pressure after 2 hours is 800 mm, the value of rate constant is:

(A) 4.63 hr^{-1}

(B) 0.346 hr^{-1}

(C) 0.463 hr^{-1}

(D) 3.46 hr^{-1}

56. Which of the following statements is correct?

(A) D-Glucose and D-Fructose do not reduce Fehling's solution.

(B) Glycogen and dextrin are reducing sugars.

(C) α-D(+) glucopyranose is an example of hemiketal.

(D) Reduction with Na-Hg/H_2O, D-Glucose gives D-Sorbitol.

57. Glycerol can be oxidized to the lowest carbohydrate by using:

(A) $Bi(NO_3)_3$

(B) dil. HNO_3

(C) conc. HNO_3

(D) Fenton's reagent

58. 5 moles of an ideal mono atomic gas are heated from 300 K to 600 K at constant volume. The entropy change during the process is:

(A) 6.30 cal/K

(B) 10.39 cal/K

(C) 0.63 cal/K

(D) 3.46 cal/K

59. Which of the following alloys does not contain copper?

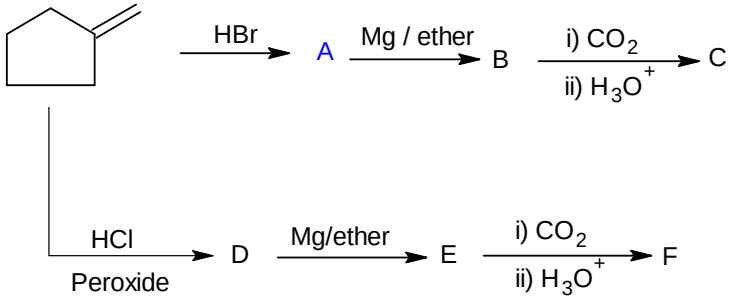
(A) German silver

(B) Magnelium

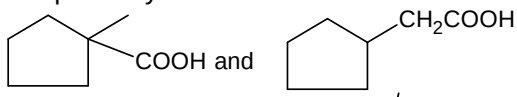
(C) Gun metal

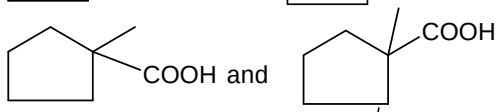
(D) Bronze

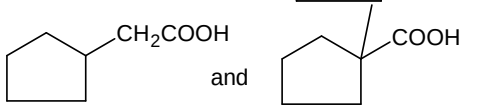
60. In the following reaction sequence:

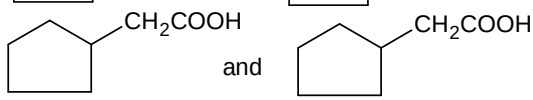


C and F respectively are

(A) 

(B) 

(C) 

(D) 

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Number of six digit numbers whose sum of the digits is 49 are
(A) $^{12}C_5$
(B) 6
(C) $^{10}C_5$
(D) 1200

62. A square matrix A satisfies the relation $A^2 = 2I - A$, where I is the identity matrix, $A^5 = KI + PA$, then P + K is equal to (where $P, K \in R$)
(A) 6
(B) –6
(C) –1
(D) 1

63. $\int \frac{1-\cos x - x\sin x}{x^2+1-2x\sin x}dx = \tan^{-1}(f(x)) + c$, then f(n) is
(A) continuous at x = 0
(B) an odd function
(C) an even function
(D) $f\left(\frac{\pi}{2}\right) = 1$

64. If f(x) = ||sin(|x| – 2)| – 3|, then
(A) f(x) is continuous but not differentiable at x = 4
(B) f(x) is discontinuous at x = 4
(C) f(x) is differentiable at x = 4 and f′(4) = cos 2
(D) f(x) is differentiable at x = 4 and f′(4) = –cos 2

65. $\frac{\sin\theta}{\cos\theta} + \frac{\sin 2\theta}{\cos^2\theta} + \frac{\sin 3\theta}{\cos^2\theta} + ... + \frac{\sin 6\theta}{\cos^6\theta}$ is equal to
(A) $\cot\theta + \frac{\cos 7\theta}{\sin\theta\cos^6\theta}$
(B) $\cot\theta - \frac{\cos 7\theta}{\sin\theta\cos^6\theta}$
(C) $\frac{\cos 7\theta}{\sin\theta\cos^6\theta} - \cot\theta$
(D) 21 tan θ

66. $f'(x) = f(x) + \int_0^1 f(x)dx$ and f(0) = 1, then $\int f(x)dx$ is equal to
(A) $\frac{2}{3-e}e^x + \left(\frac{3-e}{1-e}\right)x + c$
(B) $\frac{2}{3-e}e^x + \left(\frac{1-e}{3-e}\right)x + c$

(C) $\frac{2}{2-e}e^x + \left(\frac{e}{e-2}\right)x + c$

(D) $\frac{3}{2-e}e^x + \left(\frac{1-e}{2-e}\right)x + c$

67. In ΔABC, the sides a, b, c are the roots of the equation $x^3 - 12x^2 + 47x - 60 = 0$. Then $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$ is equal to

(A) $\frac{7}{15}$

(B) $\frac{8}{15}$

(C) $\frac{12}{5}$

(D) $\frac{5}{12}$

68. If the radii of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 8x + 10y + 32 = 0$ are increasing uniformly with respect to time as 0.2 unit/sec and 0.3 unit/sec respectively. If t_1 and t_2 are the time when they touch each other, then $|t_1 - t_2|$ is equal to

(A) 116 seconds
(B) 76 seconds
(C) 68 seconds
(D) 108 seconds

69. If $4^{\cos^2 x + \cos^4 x + \cos^6 x + ... + \infty}$ satisfies the equation $t^2 - 20t + 64 = 0$, then number of ($x \neq n\pi$) values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ are

(A) 2
(B) 4
(C) 6
(D) 18

70. The vertex A of an equilateral triangle ABC is (5, 4) and the sides AB and BC are the members of the family of lines $\ell x + my + n = 0$. If ℓ, m, n are in A.P., then area of triangle ABC is

(A) $\frac{\sqrt{39}}{2}$

(B) $10\sqrt{3}$

(C) $4\sqrt{3}$

(D) $13\sqrt{3}$

71. Domain and range of the function $f(x) = \sqrt{\sin^{-1}(3x) + \frac{\pi}{3}}$ is $\left[\frac{a}{\sqrt{3}}, \frac{b}{3}\right]$ and $\left[c, d\sqrt{5\pi}\right]$ respectively, then 2a + b + c + 6d is equal to

(A) 1
(B) $2\sqrt{3}$
(C) $\sqrt{6}$
(D) none of these

72. Let $f_k(\theta) = \sum_{k=0}^{n} \frac{1}{4^k} \sin^4\left(2^k\theta\right)$. Then which of the following is correct?

(A) $f_4\left(\frac{3\pi}{2}\right) = 2$

(B) $f_4\left(\frac{3\pi}{2}\right) = 1$

(C) $f_5(\pi) = 1$

(D) $f_5\left(\frac{\pi}{8}\right) = \frac{1}{\sqrt{2}}$

73. Let A = {x: |x – 1| < 3} and B = {x : x^2 – 2ax + a^2 – 4 ≥ 0} be two sets.
If A ∩ B = {x : –2 < x ≤ 1}, then the possible integral value of a is
(A) 7
(B) 8
(C) 3
(D) 1

74. Let t_n be n^{th} term of a geometric progression with common ratio r. If r < 0, $\frac{t_6}{t_8} = \frac{1}{4}$ and $t_3 + t_7 = 100$ and $t_1 = \frac{a}{b}$, then the value of a + b is (a and b are coprime numbers)
(A) 37
(B) 83
(C) 50
(D) 42

75. Consider a triangle PQR and p, q, r denote the length of the sides opposite to vertices P, Q and R respectively. If p, q, r are in A.P. and $x^2 + 3x + 5 = 0$ and $3x^2 + px + r = 0$ have a common root. Then the sum of inradius and circumradius of triangle PQR is
(A) 7
(B) $\frac{15}{2}$
(C) $\frac{21}{2}$
(D) $\frac{17}{2}$

76. Let $f(n) = \sum_{k=-n}^{n} \left(\cot^{-1}\left(\frac{1}{k}\right) - \tan^{-1}k\right)$, k ≠ 0. If the value of $\sum_{n=2}^{20} \left(f(n) + f(n-1)\pi\right)$ is mπ, then the vale of m is
(A) 100
(B) 98
(C) 99
(D) 117

77. If $\sum_{r=0}^{n} \left(\frac{r+2}{r+1}\right) {}^nC_r = \frac{2^{17}-1}{14}$, then the value of n is equal to
(A) 14
(B) 13
(C) 10
(D) none of these

78. Let $L = \prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M = \prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$; $N = \prod_{n=1}^{\infty}\frac{\left(1+n^{-1}\right)^2}{1+2n^{-1}}$, then the value of $L^{-1} + M^{-1} + N^{-1}$ is

(A) 0
(B) 1
(C) 8
(D) 10

79. Given that z_1, z_2 and z_3 are complex numbers with $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1z_2z_3 = 1$, then $|(z_1 + 2)(z_2 + 2)(z_3 + 2)|$ is equal to

(A) 14
(B) 15
(C) 20
(D) 9

80. A restaurant has 5 menu items. If the first 4 customers each choose one menu item at random, the probability that the 4th customer order a previously unordered item is $\frac{m}{n}$, where m and n are relatively prime positive integers. Then m + n is equal to

(A) 179
(B) 189
(C) 207
(D) none of these

81. Let circles C_1 and C_2 on Argand plane be given $|z + 2| = 4$ and $|z - 1| = 10$ respectively. If a variable circle $|z - z_0| = r$ be inside circle C_2 such that it touches C_1 externally and C_2 internally, then locus of z_0 is a conic whose eccentricity is

(A) $\frac{1}{2}$
(B) $\frac{3}{10}$
(C) $\frac{3}{8}$
(D) $\frac{3}{14}$

82. Let f: R $\to$ R be a differentiable function such that f(1) = 1, f(2) = 20, f(–4), f′(0) = 0 and $f(x + y) = f(x) + f(y) + 3xy(x + y) + bxy + c(x + y) + 4 \ \forall\ x, y \in R$ where b, c are constants, then number of solutions of the equation $f(x) = x^3 + 4e^x$ is equal to

(A) 0
(B) 1
(C) 2
(D) 3

83. Consider a tetrahedron ABCD with AB = 12, CD = 6. The shortest distance between the skew lines AB and CD is 8 and the angle between them is $\frac{\pi}{6}$. If the volume of tetrahedron is v, then $\frac{v}{12}$ is

(A) 5
(B) 8
(C) 4
(D) 2

84. Let f be a differentiable function on R and satisfying $f(x) = -\left(x^2 - x + 1\right)e^x + \int_0^x e^{x-y} f'(y)\,dy$. If f(1) + f′(1) = ke, where k ∈ N, then k is equal to
(A) 0
(B) 3
(C) 6
(D) 4

85. The radius of the circle passing through the vertices of the triangle formed by a pair of straight lines $23x^2 + 26xy + 5y^2 = 0$ and $3x + y - 1 = 0$ is
(A) 5
(B) $\sqrt{13}$
(C) $\sqrt{29}$
(D) $2\sqrt{13}$

86. The number of integral values of a for which three distinct chords of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$ passing through the point $\left(20a, -\frac{a^2}{2}\right)$ are bisected by the parabola $y^2 = 4ax$ is
(A) 14
(B) 8
(C) 6
(D) 20

87. Let f(x) be a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t)\,dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{4}$. Then the numbers of values of x in [0, 4π] for which f(x) = cosec x is
(A) 4
(B) 3
(C) 1
(D) 0

88. The director circle of an ellipse is $x^2 + y^2 - 16x = 0$. One end of the major axis is (3, 5), then one of its focus is
(A) $\left(8 + 3\sqrt{3}, \sqrt{2}\right)$
(B) $\left(8 - 3\sqrt{2}, 3\sqrt{2}\right)$
(C) $\left(\frac{37}{5}, \frac{3}{5}\right)$
(D) $\left(\frac{37}{5}, \frac{-3}{5}\right)$

89. If $\theta_1, \theta_2, \theta_3 \in [0, 3\pi]$, then the number of ordered triplets $(\theta_1, \theta_2, \theta_3)$ which satisfy $(1 + \text{cosec}^4\,\theta_1)(2 + \cot^4\,\theta_2)(4 + \sin 4\theta_3) \leq 12 \sin^2\,\theta_1$ are
(A) 18
(B) 36
(C) 72
(D) 48

90. If $I_1 = \int_{-2017}^{2019} \frac{dx}{\left(5+2x-x^2\right)\left(1+e^{3x-3}\right)}$ and $I_2 = \int_{-2017}^{2019} \frac{dx}{\left(5+2x-x^2\right)}$, then $\frac{I_1}{I_2}$ is equal to

(A) $\frac{1}{2}$

(B) 2

(C) 0

(D) none of these